



MINOR PROJECT

SYNOPSIS REPORT

For

AI-Based Sports Platform for Analysis and Ranking

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Synopsis Report

1. Project Title

AI-Based Sports Platform for Analysis and Ranking

2. Abstract

This project presents an AI-powered web platform designed to democratize sports talent assessment across India by enabling remote, accurate evaluation of athletic performance. The platform addresses the critical challenge faced by rural athletes who lack access to standardized Sports Authority of India (SAI) fitness testing facilities. By leveraging cutting-edge technologies including MediaPipe for real-time pose detection, TensorFlow.js for browser-based machine learning, and computer vision algorithms, the system allows athletes to record fitness test videos using smartphones and receive instant AI-driven performance analysis.

The platform implements five standard SAI fitness tests: pushups, sit-ups, vertical jump, shuttle run, and endurance run. Using pose estimation and joint angle calculations, the system automatically counts repetitions, measures performance metrics, and compares results against age and gender-based benchmarks. The solution employs a Progressive Web Application (PWA) architecture with React.js frontend, Python Flask backend, MongoDB database, and MinIO for video storage, ensuring accessibility across devices without requiring app installation.

Key innovations include on-device AI processing for privacy preservation, offline capability for low-bandwidth areas, and anti-cheating mechanisms through timestamp verification and pose consistency analysis. The platform features a secure dashboard for SAI officials to review submissions, identify talent pools, and export athlete profiles. With an estimated development timeline of three months following Agile methodology, this project aims to transform sports talent identification by making standardized assessment accessible to every athlete regardless of geographic or economic constraints.

3. Introduction

India, with its vast population of 1.4 billion, possesses immense potential for discovering and nurturing athletic talent. However, the current sports talent identification system faces significant challenges in reaching athletes beyond urban centers. The Sports Authority of India (SAI) has established standardized fitness test protocols to evaluate athletic potential, but these assessments require physical presence at designated testing centers, creating a substantial barrier for rural and economically disadvantaged youth.

Traditional talent identification relies on manual observation by coaches and officials, which is resource-intensive, geographically constrained, and susceptible to subjective bias. Athletes from remote villages often remain undiscovered simply because they cannot afford to travel to testing centers in cities. This systemic inefficiency means India potentially misses out on exceptional talent that could represent the nation at international competitions, including the Olympics.

Recent advances in artificial intelligence and computer vision have opened new possibilities for remote performance assessment. Technologies such as MediaPipe (developed by Google) enable real-time pose estimation from video footage, while TensorFlow.js allows machine learning models to run directly in web browsers. These innovations make it technically feasible to conduct standardized fitness assessments remotely using nothing more than a smartphone camera.

This project directly responds to Problem Statement 25073 issued by the Sports Authority of India through Smart India Hackathon 2025, which explicitly calls for innovative AI-based solutions to enable remote sports talent assessment and ranking. The platform aims to create equal opportunity for athletes regardless of location, reducing per-athlete assessment costs by approximately 95% while expanding geographic coverage from current urban centers to all 740 districts of India.

4. Problem Statement

The core problem this project addresses is the lack of accessible, standardized sports talent assessment infrastructure for athletes in rural and remote areas of India. Current challenges include:

Geographic Accessibility Barrier: Athletes must physically travel to designated SAI testing centers predominantly located in urban areas. For a young athlete in rural Uttarakhand, Bihar, or Madhya Pradesh, reaching the nearest testing facility might require traveling hundreds of kilometers with prohibitive costs for families with limited financial resources.

Limited Assessment Capacity: The current infrastructure can evaluate only a small fraction of India's youth population. With finite numbers of trained assessors and testing facilities, SAI can conduct perhaps tens of thousands of assessments annually, whereas identifying truly exceptional athletic talent from a population of over 400 million youth would ideally require evaluating millions of candidates.

Economic Constraints: Conducting in-person fitness assessments involves substantial costs including facility maintenance, equipment, personnel salaries, travel expenses, and administrative overhead. These expenses limit the frequency and geographic distribution of assessment drives, particularly in economically disadvantaged regions where the need is greatest.

Subjective Evaluation Challenges: Human observation introduces variability. Different coaches might assess the same pushup performance differently based on viewing angle, attention level, or personal judgment criteria. This subjectivity can lead to inconsistent standards across different testing venues.

Data Fragmentation: Existing assessment records are often maintained in disparate systems without centralized integration, making it difficult to identify talent patterns across regions, track long-term athlete development, or efficiently connect promising

athletes with appropriate training programs.

The fundamental challenge is creating a scalable, accessible, objective, and cost-effective system for identifying athletic talent that reaches every potential athlete regardless of location or economic status, while maintaining rigorous evaluation standards comparable to professional assessment centers.

5. Literature Review

The development of AI-powered fitness assessment builds upon significant prior research and technological development in several domains:

MediaPipe Framework: Lugaresi et al. (2019) introduced MediaPipe as an open-source framework for building perception pipelines. Their pose detection model achieves 95%+ accuracy in identifying 33 body landmarks in real-time, even on mobile devices. This breakthrough makes browser-based pose estimation practical for consumer applications. The framework's efficiency stems from its use of a detector-tracker architecture that runs inference only when necessary, reducing computational overhead.

TensorFlow.js: Smilkov et al. (2019) demonstrated that complex machine learning models could execute efficiently in web browsers through WebGL acceleration. Their work enables on-device inference, addressing privacy concerns and reducing server costs. For sports applications, this means athletes' videos can be analyzed locally without uploading sensitive footage to servers.

Computer Vision in Sports: Zecha et al. (2019) explored kinematic pose rectification for performance analysis in sports, showing that pose estimation accuracy is sufficient for professional coaching feedback. Their research validated that AI systems can match human expert analysis for many standard movements.

Exercise Recognition Systems: Khurana et al. (2018) developed GymCam, demonstrating automated detection and counting of exercises in unconstrained environments. Their work showed that machine learning models could reliably distinguish between different exercises and count repetitions with >90% accuracy.

SAI Fitness Standards: The Sports Authority of India has published comprehensive fitness test protocols through their National Talent Identification and Development Scheme. These protocols specify exact procedures for measuring physical capabilities including strength (pushups, pull-ups), endurance (distance runs), agility (shuttle runs),

and power (vertical jumps). These standards, derived from testing thousands of Indian athletes, account for population-specific characteristics and serve as benchmark database foundations.

OpenCV Applications: Bradski's (2000) OpenCV library has become the de facto standard for computer vision tasks. Its robust implementations of video processing, frame extraction, and quality assessment are essential for preprocessing video inputs before pose analysis.

6. Objectives

The primary objective of this project is to create an accessible, web-based platform leveraging artificial intelligence to democratize sports talent assessment across India. The platform will enable remote evaluation of athletes' fitness levels using standardized SAI protocols, thereby expanding talent identification reach to rural and underserved populations who currently lack access to formal assessment infrastructure.

Specific objectives include:

1. **Video Recording and Upload Functionality:** Implement browser-based video recording capabilities using the MediaRecorder API, allowing athletes to capture fitness test performances directly through the web interface without requiring additional software installation. The system will provide real-time guidance on camera positioning and test procedures.
2. **AI-Based Exercise Detection and Measurement:** Develop machine learning pipelines using MediaPipe pose detection to automatically identify body landmarks, calculate joint angles, and measure performance metrics such as repetition counts, movement amplitude, and execution speed with accuracy comparable to human observers.
3. **Age and Gender-Based Performance Benchmarking:** Create a comprehensive database of fitness standards aligned with SAI protocols, segmented by age groups and gender. Implement algorithms to compare individual athlete performance against relevant benchmark cohorts and generate percentile rankings.
4. **Cheat Detection and Verification Mechanisms:** Build multi-layered verification systems including timestamp watermarking, video metadata analysis, pose consistency checking, and physics-based movement validation to detect and prevent fraudulent submissions while maintaining legitimate user experience.

5. Secure SAI Official Dashboard: Design and implement an administrative portal enabling Sports Authority of India officials to review athlete submissions, approve or flag videos, generate talent reports, filter by region and performance level, and export data for further analysis and recruitment purposes.

7. Methodology

The platform follows a three-tier architecture optimized for scalability, security, and user experience:

Presentation Layer (Frontend): Built as a Progressive Web Application using React.js framework with TailwindCSS for responsive design. This layer handles user interface, video recording via MediaRecorder API, and client-side pose detection using MediaPipe loaded through TensorFlow.js. The PWA approach enables offline functionality and app-like experience without installation requirements.

Application Layer (Backend): Implemented using Python Flask framework for its simplicity and excellent integration with machine learning libraries. This layer manages user authentication, video processing workflows, database operations, AI model inference coordination, and RESTful API endpoints for client-server communication.

Data Layer: Comprises MongoDB for flexible document storage (athlete profiles, test results, AI analysis data) and MinIO object storage for video files. MongoDB's schema-less design accommodates evolving data structures as new features are added, while MinIO provides S3-compatible, cost-effective video storage with geographic distribution capabilities.

The architecture employs asynchronous processing where appropriate: videos upload to MinIO, triggering backend analysis jobs that run independently of user sessions. Results are stored in MongoDB and pushed to clients via WebSocket connections when ready, ensuring responsive user experience even for computationally intensive operations.

Development Timeline:

The project follows a three-month development schedule aligned with academic semester constraints and Smart India Hackathon 2025 submission deadlines:

Month 1 (Setup & UI Development): Environment setup, database schema design, basic authentication system, UI scaffolding with React components, video upload functionality, and continuous integration/deployment pipeline establishment. Success criteria: Users can register, login, and upload videos.

Month 2 (AI Integration): MediaPipe pose detection integration in browser, exercise classification algorithms, repetition counting for pushups and sit-ups, vertical jump height measurement logic, benchmark database population with SAI standards, and performance comparison implementation. Success criteria: System correctly counts 90%+ of valid test repetitions.

Month 3 (Advanced Features & Deployment): Anti-cheating mechanisms (metadata validation, pose consistency), shuttle run and endurance run tracking, SAI official dashboard with filtering capabilities, data export functionality, comprehensive testing across devices and browsers, security audits, performance optimization, and production deployment. Success criteria: System handles 100 concurrent users with <2 second response times.

8. Technology Stack

The technology selection prioritizes open-source solutions, proven reliability, active community support, and alignment with project requirements:

Frontend Technologies:

React.js 18.2 serves as the component-based JavaScript library for building user interfaces, with virtual DOM optimization ensuring smooth rendering performance.

TailwindCSS 3.3 provides utility-first CSS framework for rapid UI development without writing custom CSS. MediaRecorder API enables browser-native video recording without external libraries. MediaPipe Web delivers Google's pose estimation solution compiled to WebAssembly for browser execution. TensorFlow.js 4.x enables machine learning library for browser-based neural network inference with WebGL acceleration.

Backend Technologies:

Python 3.11 provides modern Python version with performance improvements and enhanced type hinting. Flask 3.0 offers lightweight microframework for web applications with WSGI compliance. Flask-JWT-Extended handles JWT authentication with refresh token support and role-based access control. NumPy 1.24 delivers numerical computing library for efficient array operations essential for mathematical calculations. OpenCV-Python 4.8 provides computer vision library for video preprocessing including quality assessment and frame extraction. PyMongo 4.5 serves as official MongoDB driver with connection pooling. Minio-Python 7.2 offers MinIO client library providing S3-compatible object storage interface.

Data Storage:

MongoDB 6.0 provides NoSQL document database offering flexible schema, rich query language, and aggregation framework. MinIO delivers high-performance object storage with S3 compatibility, deployed on-premises for development with cloud-ready architecture for production. Redis 7.2 serves as in-memory data structure store for caching and message brokering.

Development and Deployment Tools:

Git provides version control with GitHub for collaboration. Docker enables containerization for consistent development/production environments. Nginx serves as reverse proxy and load balancer. Pytest and Jest provide testing frameworks for backend and frontend respectively.

9. Expected Outcomes

Upon successful project completion by May 2025, the following deliverables will be provided:

Functional Web Platform: A fully operational Progressive Web Application accessible via standard browsers on desktop and mobile devices, requiring no installation.

Responsive design adapting to various screen sizes with smooth transitions and animations enhancing user experience. Real-time video recording with countdown timers and form guidance. Offline functionality for viewing historical results and basic navigation.

AI-Powered Assessment System: MediaPipe-based pose detection operating at 25-30 FPS on modern devices. Automated analysis of five SAI standard fitness tests: pushups (rep counting with form quality assessment), sit-ups (60-second timed counting), vertical jump (height measurement in centimeters), shuttle run (time recording with line-crossing detection), and endurance run (distance/time tracking). Accuracy within $\pm 5\%$ of human expert evaluation validated against test dataset. Client-side processing preserving athlete privacy with server-side fallback for complex analysis requirements.

Comprehensive Database: MongoDB collections storing athlete profiles (demographics, test history, performance trends), test results with full AI analysis metadata, SAI benchmark standards (age/gender stratified), and official user accounts with role-based permissions. Indexing optimized for common query patterns. Data backup and recovery procedures. Estimated capacity: 100,000+ athlete records, 1 million+ test results.

Official Dashboard: Administrative interface for SAI officials featuring athlete submission review with video playback and AI overlay, advanced search and filtering (region, performance level, age group), talent discovery tools identifying high-potential athletes, geographic heat maps showing performance distributions, and export functionality (CSV athlete lists, PDF individual reports). Role-based access control with audit logging.

Technical Documentation: System Architecture Document, API Documentation with all endpoints, Database Schema Document, Deployment Guide with step-by-step production deployment instructions, User Manual for athletes, and Administrator Manual for officials.

Social Impact: Estimated first-year reach of 50,000 athletes assessed (10× traditional capacity). Cost savings of ₹2.5 crore versus traditional methods. Geographic coverage extending to all 740 districts (versus current ~50 cities). Approximately 500 athletes in ≥90th percentile previously undiscovered. Expected 40% female participation (versus ~25% in traditional assessments).

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